Speed Breaker and Pothole Detection and Alarm System

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***Abstract*— Road obstacles such as speed bumps and potholes are significant contributors to accidents in the World, where around 60,000 accidents were reported due to potholes between 2014 and 2022 alone. Advancing safety technologies have been implemented such as ADAS (Advanced Driver Assistance System). Although their effectiveness has been proven through real-life scenarios, ADAS has trouble detecting smaller objects such as potholes or speed breakers and they are inly present in high-end luxury vehicles. Measures are taken using machine learning models integrated with camera and ultrasonic sensor to reduce such accidents. The system consists of a raspberry Pi 4 Model B which is connected to OV5647 video camera, US100 Ultrasonic sensor and a PCB mounted Buzzer. It is powered by two Samsung 18650 batteries. The Raspberry Pi stores the YOLOv5s machine learning model. The system monitors the road environment at around 85% accuracy level and sends instant alerts through its sensor network. Our solution offers a practical cost-effective way to increase road safety performance**

***Keywords*— YOLOv5, Deep Learning, Raspberry Pi, Pothole Detection, Speed Breaker Detection, Road Hazard Identification, Real-Time Hazard Detection and Alert, Image Processing.**

I. INTRODUCTION

Road safety matters most since potholes and unmarked speed breakers generate substantial danger for both vehicle drivers and their passengers. These hazards create dangerous incidents and deaths because most systems lack accurate ways to alert the driver. Budget and mid-range vehicles are at greater risk due to this lower safety ratings. Our research aims to build an affordable system that detects the locations of potholes and unmarked speed breaker using camera and ultrasonic sensors to give real-time alerts to the driver of the vehicle. In India, between the period 2014-2022, potholes led to more than 60,000 vehicle accidents and between the period 2014-2016, speed breakers caused over 30,000 crashes. Over 31000+ fatalities were caused in this period. These advanced safety features exist today only in certain luxury vehicles such as those from Mercedes-Benz and Range Rover because higher deployment costs. This system would make this important safety feature available to all vehicle types while keeping costs low. If the problem is unaddressed, the number of such accidents and fatalities will continue to rise. This will lead to increased vehicle repair and maintenance cost, traffic congestion due to accidents, increased pollution levels, and even certain health problems such as spinal compression and lower back pains. The system delivers real-time alerts that will help reduce traffic risks and improve driver and passenger safety. This research focuses on drivers and passengers who own private or commercial vehicles plus municipal officials and business partners who work in vehicle production and navigation. The system monitors the road environment at around 85% accuracy level and sends instant alerts through its sensor network. The project scope is to test technology in urban areas but it excludes the use of a GPS module, data storage and physical repair of potholes. The project operates under constraints such as a budget limit of ₹10,000 and a six-month completion timeline. Our solution offers a practical cost-effective way to increase road safety performance.

II. LITERATURE SURVEY

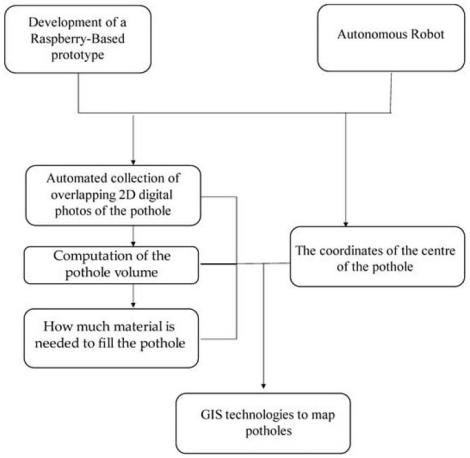


Fig. 1 Block Diagram of ref [1]

[1] This research displays a low-cost, robotic system used for three-dimensional pothole mapping using a Raspberry Pi microprocessor. This system captures images appended with geographical co-ordinates using a remote-controlled robot, that moves in a circular path around the potholes. The collected frames are processed to generate a 3D model which estimates pothole volume to repair efficiently. Hardware

includes a Raspberry Pi 4B microprocessor, a GPS module, and a camera, with Python scripts which automate image capture and geotagging. QGIS software maps the pothole locations and requirements for repair. Lab and field tests confirm accuracy of the system in real-world conditions. This approach enhances road safety while minimizing worker risk.

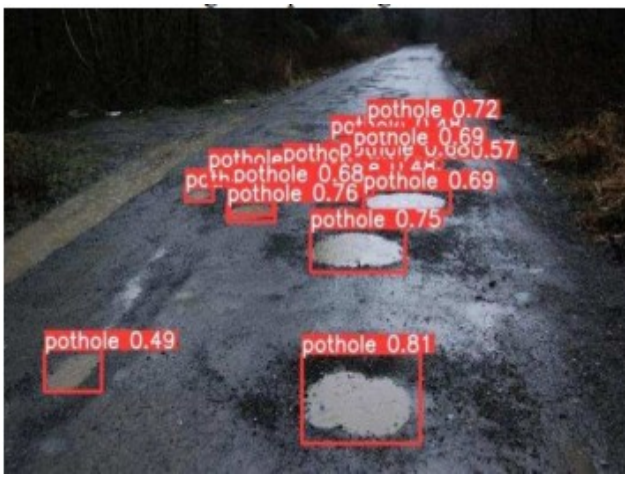


Fig. 2 Output of ref [2]

[2] The presence of potholes simultaneously creates two severe risks for road users especially vehicle owners. Standard inspection routines show poor efficiency while the planet faces crisis situations including COVID-19. The introduced system uses a deep learning method built using YOLOv5 technology for real-time detection of potholes. The system leverages YOLOv5's CSPDarknet backbone, PANet neck, and detection head. A distinct dataset (comprising wet and dry conditions with different types of potholes) received annotation for training purposes. Real-time identification coupled with size estimation provides maintenance teams precise information to prioritize road maintenance efforts. The YOLOv5 outperformed both YOLOv3 and YOLOv4 in terms of accuracy rate and training duration (15 minutes) as well as memory consumption. The detection system capabilities extend to finding potholes in captured images alongside real-time video streams. YOLOv5 provides an automated detection system for potholes through efficient processing alongside scalable orientation capabilities. Future researchers should focus on improving depth assessment functions in addition to achieving Internet of Things (IoT) capability enhancement for automatic smart road maintenance operations.

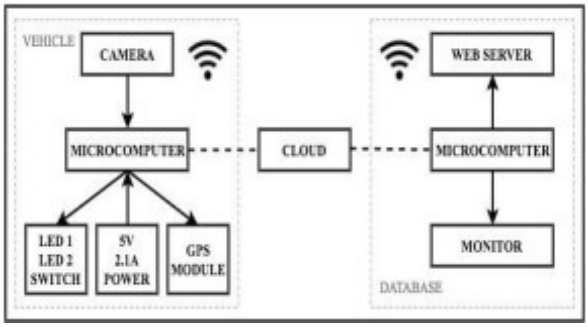


Fig. 3 Block Diagram of ref [3]

[3] Potholes function as a significant cause of traffic accidents throughout developing nations including the Philippines. Present-day methods to detect potholes require either expensive smartphone-based devices or experience challenges in proving their functionality in real-time use. The proposed low-cost system implements a real-time pothole sensing and reporting system through a combination of Raspberry Pi microcontroller and camera module and GPS unit. The system records road images which then undergo processing with OpenCV-based algorithms for pothole detection through edge detection and contour analysis. GPS tagging serves to map exact locations and data transmission happens through Wi-Fi which stores information in Dropbox as its temporary database. Ideal testing environment demonstrated 100% precise results, yet real application testing produced 87.45% accuracy because puddles combined with reflections caused incorrect positive signals. The system operates at eight frames per second and performs responses at 0.1244 seconds. Large-scale implementation becomes attainable because this system remains affordable while being portable and smartphone-independent even though its accuracy faces minor adjustments.

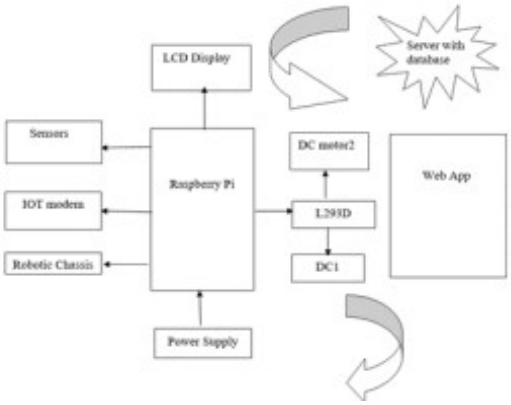


Fig. 4 Block Diagram of ref [4]

[4] Road deterioration raises the risk of collisions so automatic detection methods work together with safety protocols to deliver solutions. An IoT-based system emerges in this research to pinpoint both potholes and speed breakers while warning drivers and supporting civil authorities. The system consists of: Microcontroller Module: The system

combines ultrasonic sensors with GPS to detect potholes. Server Module: The cloud receives data inputs and keeps them in a secure location. Mobile Application Module: Sends real-time visual/audio alerts within 100m. The system based on IoT technology operates with low costs to deliver enhanced road safety standards. More enhancements will focus on combining voice warnings with maps navigation systems.

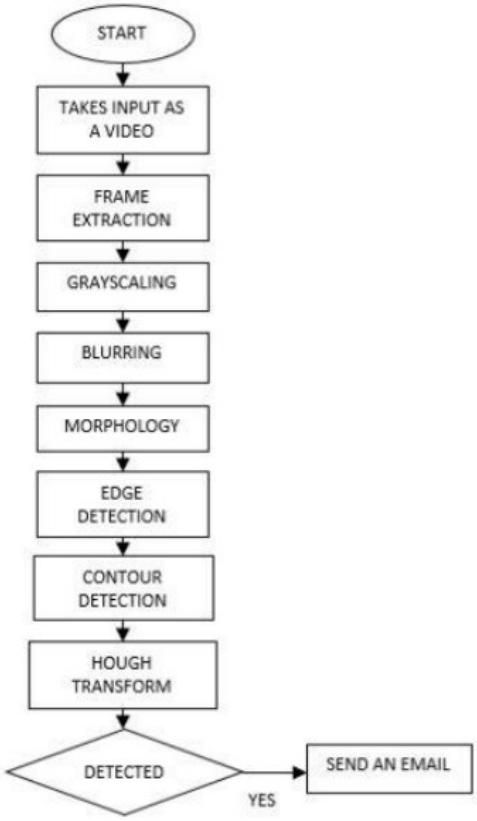


Fig. 5 Block Diagram of ref [5]

[5] In India, potholes which are caused due to poor construction and inadequate drainage, cause large number of accidents, especially at night and in foggy conditions. Fixing the roads are an ideal solution but is set back by corruption, poor planning, and time constraints. Hence, an alternative solution is real-time pothole detection and avoidance of it. This system captures videos of road conditions using a Raspberry Pi connected with a camera, that extracts frames, and processes them using grayscale conversion and reduction of noise (Gaussian filter). Otsu's method is used for image analysis, along with morphological operations for defining size of pothole and edge detection. Using InceptionResNetV2 and DenseNet201 models, an accuracy of 89.66% is achieved, which has greater accuracy other models. The low-cost implementation on Raspberry Pi guarantees pothole detection, preventing vehicle damage and enhancing road safety.

III. METHODOLOGY

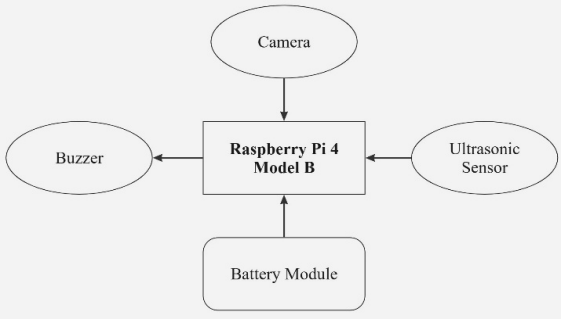


Fig. 6 Proposed System Block Diagram

A. Overview of the System

The system uses YOLOv5s, an open-source machine learning model implemented on the Raspberry Pi 4 Model B (2GB RAM version). Figure 6 shows the shows the basic block diagram of the system. System includes US-100 Ultrasonic Sensor which is used to measure the distance, height or depth of the speed breaker or pothole, Raspberry Pi 4 Model B is used as the processor containing the pre-trained machine learning model which is fine-tuned over a dataset from Kaggle which is free to use, OV-5647 Camera Module used for sending the frames to the model for real world vision., Samsung INR18650-35e 3500MAh (2c) Li-Ion cell in a battery Module provides 5v/3A output to the system components and PCB Mounted Active Buzzer is mounted on the model car for alerting user of the obstacle.

B. Training and Building of the system

The Raspberry Pi is used as microprocessor which receives the data from camera OV5647 module and the ultrasonic sensor US100. The detection process is implemented with the pre-trained YOLOv5s machine learning model which is fine-tuned on a labelled dataset from Kaggle of 1440 images which includes both marked and unmarked speed breakers and potholes. The dataset is split into training (80%) and validation (20%) with applied data augmentation for generalization. Model was trained using Stochastic Gradient Descent (SGD) and optimized using Mean Average Precision (mAP), precision and recall metrics over 50 epochs. The best model was then applied over the Raspberry Pi Microprocessor for detection. One of the GPIO pins of the Raspberry Pi is connected to an active buzzer to give real-time alerts upon detection. The Raspberry Pi receives power from a battery module consisting of two Samsung 18650 2c batteries of 3500mAH each which is sufficient to provide power to the system for approximately 3 hours. The battery module provides power of 5v 3A, which is optimal to run Raspberry Pi along with the sensors and buzzer.

C. Process Flow

The Raspberry Pi module is responsible for the decision-making based on inputs from the OV5647 camera module and the US100 ultrasonic sensor. The system operates using a YOLOv5s machine learning model, trained to detect obstacles

such as speed breakers and potholes in real-time. When the system is powered, the Raspberry Pi initializes the modules, including the OV5647 camera, US100 ultrasonic sensor, and the YOLOv5s object detection model. It then continuously captures video frames from the OV5647 camera module, providing real-time image data to the model for analysis. As each frame is processed, the YOLOv5s model detects obstacles such as speed breakers or potholes. If an obstacle is identified with a score above a predefined threshold, the model outputs a 'true' signal through the Raspberry Pi. If no obstacle is detected, the system continues capturing and analysing new frames. Upon detecting an obstacle, the Raspberry Pi activates the US100 ultrasonic sensor to measure the distance, height, or depth of the detected object. The ultrasonic sensor will provide additional confirmation by distinguishing between a raised object, such as a speed breaker, and a depression, such as a pothole, based on depth and height parameters. If the sensor's measurements match the expected characteristics of an obstacle, the Raspberry Pi validates the detection. Once an obstacle is confirmed, the system activates the buzzer to alert the user. After passing the obstacle, the system restarts the sequence, continuing to capture frames and analyse the environment in real time. This ensures that any new or missed obstacles are detected as the system operates continuously.

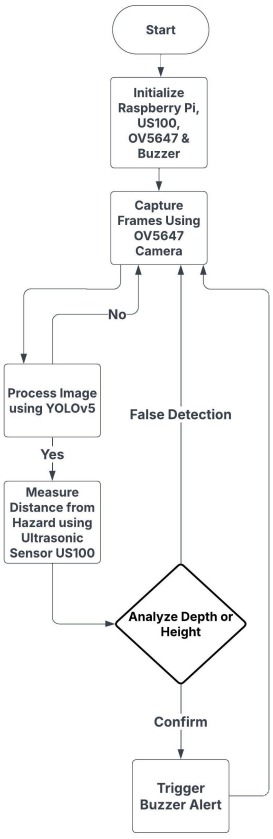


Fig. 7 Process Flow Chart

IV. Future Scope

Our speed breaker and pothole detection and alarm system currently provides improved road safety by detecting speed breakers and potholes and giving real-time alerts. In the future, such systems have the potential for several improvements to improve accuracy, efficiency, and scalability.

The possible enhancements and upgrades that can be done are internet-based, such as cloud-based data storage and analysis for the government to take action to repair potholes, or physically based integrations such as the use of better and more advanced sensor technology like Lidar to increase the distance of measurement, integration with navigation maps, and self-driving vehicles. There can also be machine learning model improvements, such as the use of machine learning models with higher precision and accuracy, the creation of a large dataset of road conditions for improvement of the model, and the expansion to poor road conditions such as rural areas and off-roads.

V. Conclusions

Through this research, it can be concluded that the proposed system could possibly be a lifesaver. If such a system is leveraged at a larger scale in a vehicle, the number of road casualties will reduce drastically. This will save lives of both, the people in the vehicle and outside. Such a system if upgraded with other features such as a GPS locator may also help governments in locating potholes.

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